

MATHEMATICS

TARGET : JEE- Advanced

CAPS-23

CIRCLE

SCQ (Single Correct Type) :

- Three concentric circles of which the biggest is $x^2 + y^2 = 1$, have their radii in A.P. If the line $y = x + 1$ cuts all the circles in real and distinct points. The interval in which the common difference of the A.P. will lie is :
 (A) $\left(0, \frac{1}{4}\right)$ (B) $\left(0, \frac{1}{2\sqrt{2}}\right)$ (C) $\left(0, \frac{2-\sqrt{2}}{4}\right)$ (D) none of these
- A circle of constant radius 'r' passes through origin O and cuts the axes of coordinates in points P and Q, then the equation of the locus of the foot of perpendicular from O to PQ is :
 (A) $(x^2 + y^2)(x^{-2} + y^{-2}) = 4r^2$ (B) $(x^2 + y^2)^2(x^{-2} + y^{-2}) = r^2$
 (C) $(x^2 + y^2)^2(x^{-2} + y^{-2}) = 4r^2$ (D) $(x^2 + y^2)(x^{-2} + y^{-2}) = r^2$
- A pair of tangents are drawn from a point P to the circle $x^2 + y^2 = 1$. If the tangents make an intercept of 2 units on the line $x = 1$, then the locus of P is _____.
 (A) a straight line (B) a pair of lines (C) a parabola (D) a hyperbola
- A circle with centre at the origin and radius equal to a meets the X axis at the points A(-a, 0) and B(a, 0). P(α) and Q(β) are two points on this circle so that $\alpha - \beta = 2\gamma$, where γ is a constant. The locus of the point of intersection of AP and BQ is _____.
 (A) $x^2 - y^2 - 2ay \tan \gamma = a^2$ (B) $x^2 + y^2 - 2ay \tan \gamma = a^2$
 (C) $x^2 + y^2 + 2ay \tan \gamma = a^2$ (D) $x^2 - y^2 + 2ay \tan \gamma = a^2$
- Let the lines $y - 2 = m_1(x - 5)$ and $(y + 4) = m_2(x - 3)$ intersect at right angles at a point P, where m_1 and m_2 are parameters. If the locus of P is $x^2 + y^2 + gx + fy + 7 = 0$, then the value of $(f - g)$ equals _____.
 (A) 1 (B) 2 (C) 8 (D) 10
- The circle, which passes through the points of intersection of the circles $x^2 + y^2 - 4x - 6y + 12 = 0$ and $x^2 + y^2 - 8x + 12y + 50 = 0$, and also passes through the origin, is _____.
 (A) $19x^2 + 19y^2 - 52x - 222y = 0$ (B) $19(x^2 + y^2) - 2(34x + 111y) = 0$
 (C) $19(x^2 + y^2) - 117x + 26y = 0$ (D) such circle does not exist
- Let P(α, β) be a point in the first quadrant. Circles are drawn through P touching the coordinate axes. The relation between α and β , for which two circles are orthogonal, is _____.
 (A) $\alpha^2 + \beta^2 = 4\alpha\beta$ (B) $(\alpha + \beta)^2 = 4\alpha\beta$
 (C) $\alpha^2 + \beta^2 = \alpha\beta$ (D) $\alpha^2 + \beta^2 = 2\alpha\beta$
- The equation of circum-circle of a ΔABC is $x^2 + y^2 + 3x + y - 6 = 0$. If A = (1, -2), B = (-3, 2) and the vertex C varies then the locus of ortho-centre of ΔABC is a
 (A) Straight line (B) Circle (C) Parabola (D) Ellipse



9. Let AB be any chord of the circle $x^2 + y^2 - 4x - 4y + 4 = 0$ which subtends an angle of 90° at the point (2,3) then the locus of the midpoint of AB is a circle whose centre is
- (A) (1, 5) (B) $\left(1, \frac{3}{2}\right)$ (C) $\left(1, \frac{5}{2}\right)$ (D) $\left(2, \frac{5}{2}\right)$
10. P and Q are two points on a line passing through (2, 4) and having slope m. If a line segment AB subtends a right angle at P and Q where A $\equiv$ (0, 0) and B $\equiv$ (6, 0), then range of m is
- (A) $\left(\frac{2-3\sqrt{2}}{4}, \frac{2+3\sqrt{2}}{4}\right)$ (B) $\left(-\infty, \frac{2-3\sqrt{2}}{4}\right) \cup \left(\frac{2+3\sqrt{2}}{4}, \infty\right)$
- (C) (-4, 4) (D) $(-\infty, -4) \cup (4, \infty)$

MCQ (One or more than one correct) :

11. If $a\ell^2 - bm^2 + 2d\ell + 1 = 0$, where a, b, d are fixed real numbers such that $a + b = d^2$, then the line $\ell x + my + 1 = 0$ touches a fixed circle :
- (A) which cuts the x-axis orthogonally
- (B) with radius equal to b
- (C) on which the length of the tangent from the origin is $\sqrt{d^2 - b}$
- (D) none of these.
12. Let A, B, C, D lie on a line such that $AB = BC = CD = 1$. The points A and C are also joined by a semicircle with AC as diameter and P is a variable point on this semicircle such that $\angle PBD = \theta$, $0 \leq \theta \leq \pi$. Let R is the region bounded by arc AP, the straight line PD and line AD.
- (A) The maximum possible area of region R is $\frac{2\pi + 3\sqrt{3}}{6}$
- (B) If 'L' is the perimeter of region 'R', then L is equal to $3 + \pi - \theta + \sqrt{5 - 4\cos\theta}$
- (C) The maximum possible area of region R is $\frac{2\pi - 3\sqrt{3}}{6}$
- (D) If 'L' is the perimeter of region 'R', then L is equal to $3 + \pi - \theta + \sqrt{5 + 4\cos\theta}$

Numerical based Questions :

13. The axes are translated so that the new equation of the circle $x^2 + y^2 - 5x + 2y - 5 = 0$ has no first degree terms and the new equation $x^2 + y^2 = \frac{\lambda}{4}$, then find the value of λ .
14. A line meets the co-ordinate axes in A and B. A circle is circumscribed about the triangle OAB. If d_1 and d_2 are the distances of the tangent to the circle at the origin O from the points A and B respectively and diameter of the circle is $\lambda_1 d_1 + \lambda_2 d_2$, then find the value of $\lambda_1 + \lambda_2$.
15. Find the number of integral points which lie on or inside the circle $x^2 + y^2 = 4$.
16. Find number of values of 'c' for which the set,
- $\{(x, y) | x^2 + y^2 + 2x \leq 1\} \cap \{(x, y) | x - y + c \geq 0\}$ contains only one point is common.

17. A rhombus is inscribed in the region common to the two circles $x^2 + y^2 - 4x - 12 = 0$ and $x^2 + y^2 + 4x - 12 = 0$ with two of its vertices on the line joining the centres of the circles and the area of the rhombus is $a\sqrt{3}$ sq. units, then find the value of a .
18. Let A be the centre of the circle $x^2 + y^2 - 2x - 4y - 20 = 0$. Suppose that the tangents at the points $B(1, 7)$ & $D(4, -2)$ on the circle meet at the point C . Find the area of the quadrilateral $ABCD$.
19. If a tangent of slope $\frac{1}{2}$ of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is normal to the circle $x^2 + y^2 + 4x + 2 = 0$, then the maximum value of ab is _____.

Subjective Type Questions:

20. Find the equation of the circle passing through the points $A(4, 3)$, $B(2, 5)$ and touching the axis of y . Also find the point P on the y -axis such that the angle APB has largest magnitude.
21. Two circles, each of radius 5 units, touch each other at $(1, 2)$. If the equation of their common tangent is $4x + 3y = 10$. Find the equations of the circles.
22. The centre of the circle $S = 0$ lies on the line $2x - 2y + 9 = 0$ and $S = 0$ cuts orthogonally the circle $x^2 + y^2 = 4$. Show that circle $S = 0$ passes through two fixed points and also find their co-ordinates.
23. The lines $5x + 12y - 10 = 0$ and $5x - 12y - 40 = 0$ touch a circle C_1 of diameter 6 unit. If the centre of C_1 lies in the first quadrant, find the equation of the circle C_2 which is concentric with C_1 and cuts off intercepts of length 8 on these lines.
24. Prove that the two circles which pass through the points $(0, a)$, $(0, -a)$ and touch the straight line $y = mx + c$ will cut orthogonally if $c^2 = a^2(2 + m^2)$.
25. Show that if one of the circle $x^2 + y^2 + 2gx + c = 0$ and $x^2 + y^2 + 2g_1x + c = 0$ lies within the other, then gg_1 and c are both positive.

ANSWER KEY OF CAPS-23

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| 1. (C) | 2. (C) | 3. (C) | 4. (B) | 5. (D) |
| 6. (D) | 7. (A) | 8. (B) | 9. (D) | 10. (B) |
| 11. (AC) | 12. (AB) | 13. (49) | 14. (2) | 15. (13) |
| 16. (1) | 17. (8) | 18. (75) | 19. (1) | |
| 20. $x^2 + y^2 - 4x - 6y + 9 = 0$ OR $x^2 + y^2 - 20x - 22y + 121 = 0$, $P(0, 3)$, $\theta = 45^\circ$ | | | | |
| 21. $x^2 + y^2 + 6x + 2y - 15 = 0$; $x^2 + y^2 - 10x - 10y + 25 = 0$ | | | | |
| 22. $(-4, 4)$; $\left(-\frac{1}{2}, \frac{1}{2}\right)$ | 23. $x^2 + y^2 - 10x - 4y + 4 = 0$ | | | |
| 24. Hence proved | 25. $\therefore gg_1$ will also be > 0 | | | |